DESTINATION DURHAM · COMPUTER SCIENCE TASTER

Durham University

How a Computer *Really* Works

From a single switch to a running program in two hours

Dr Carlton Shepherd · Durham University

cs.gl/riscv-web-emulator/

About Me

- Dr Carlton Shepherd (Assistant Professor of Computer Science).
- I teach Computer Systems (1st year) and Networked Systems (2nd year).
- Areas of expertise: cyber security, device security (or: breaking and fixing things in a way that appears professionally respectable).

So… what actually IS a computer?

By the end of this session, you will hopefully have a better understanding.

THE NEXT TWO HOURS

Where we're going

1

Just switches

Binary: everything is on/off

2

The ladder

Layers from electrons to apps

3

Meet the CPU

Registers, memory, fetch–execute

4

Make it add

Step through real instructions

Break

10-15 minutes

5

Poke the machine

Change memory & registers by hand

6

C → machine code

Watch a language become steps

7

Draw with code

Paint pixels into memory

PART 1 · FIRST PRINCIPLES

It's all just switches

A computer can't store a “7”, a letter, or a photo. It only has switches: on or off – or, zeros and ones representing those states. Everything else is built upon this.

1

One switch = one bit. Eight switches = one byte.

Every switch is worth **double** the one to its right. Turn some on, add up their values — that's a number in **binary**.

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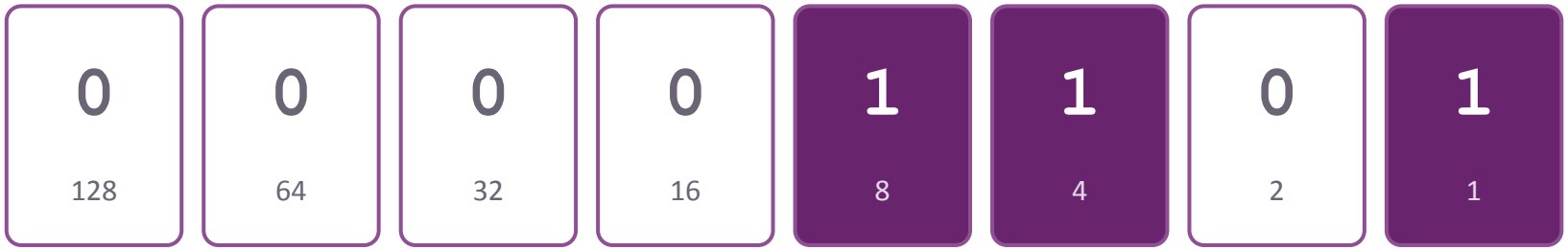

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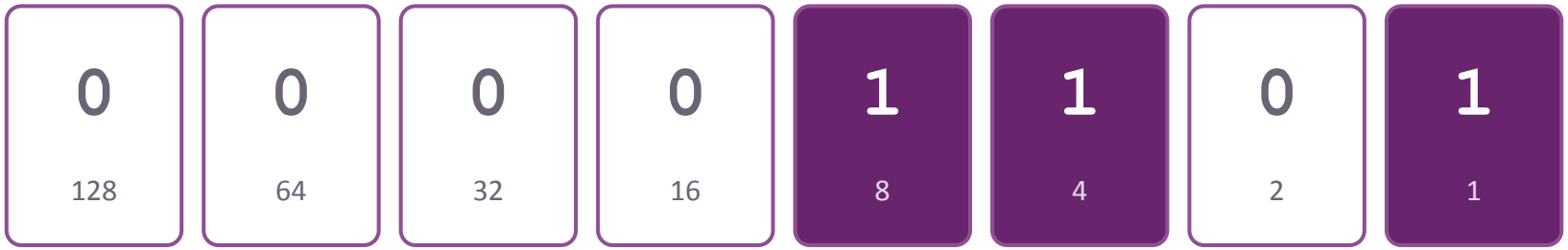

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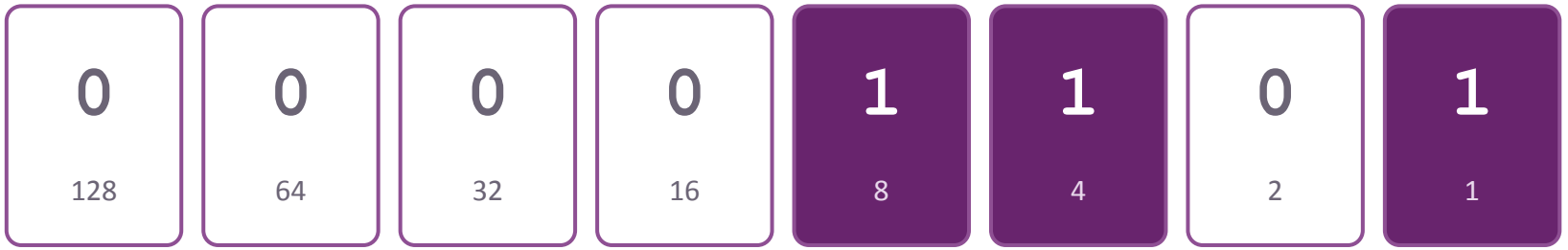

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256

different values fit in one byte — the numbers 0 to 255. That's the ceiling of 8 switches.

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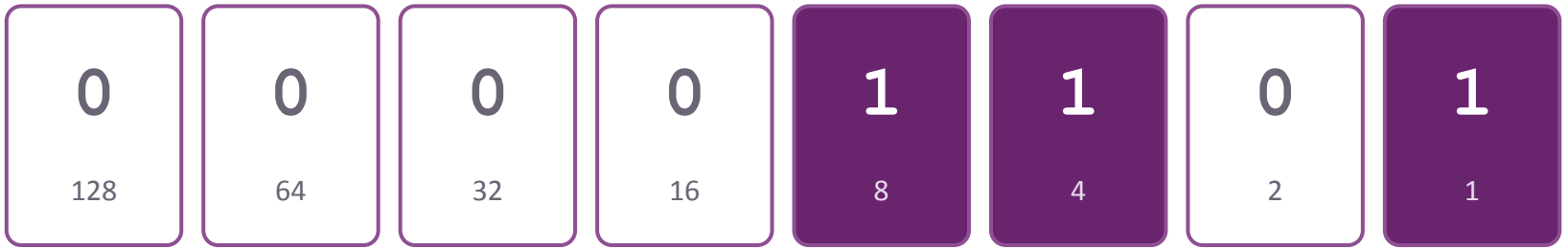

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The computer never "understands" anything; it just moves 0s and 1s insanely fast.

✋ Try it together

1 Make 13

Which switches turn do you turn on/off?

2 Make 42

Trickier.

 Go as big as you can

If we put two bytes side by side — 16 switches — what is the biggest number you can make?

Even letters are secretly numbers

There's a rule — called **ASCII** — that maps each letter to a number. The computer stores the number; we read it as a letter.

A	B	C	H	i
= 65	= 66	= 67	= 72	= 105

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65 = A = 01000001. The same bits are a number AND a letter; context decides which. That one idea is why the same machine can do maths, write essays, and show photos.

PART 2 · THE BIG PICTURE

The ladder of abstraction

Tapping an app sits at the top of a tall ladder. Let’s go down it.

2

Six rungs from electricity to apps

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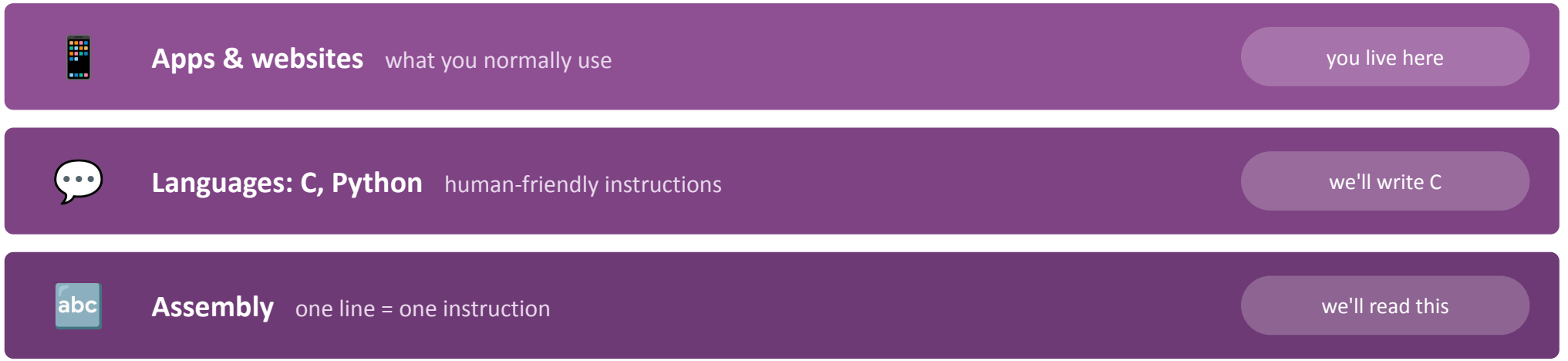

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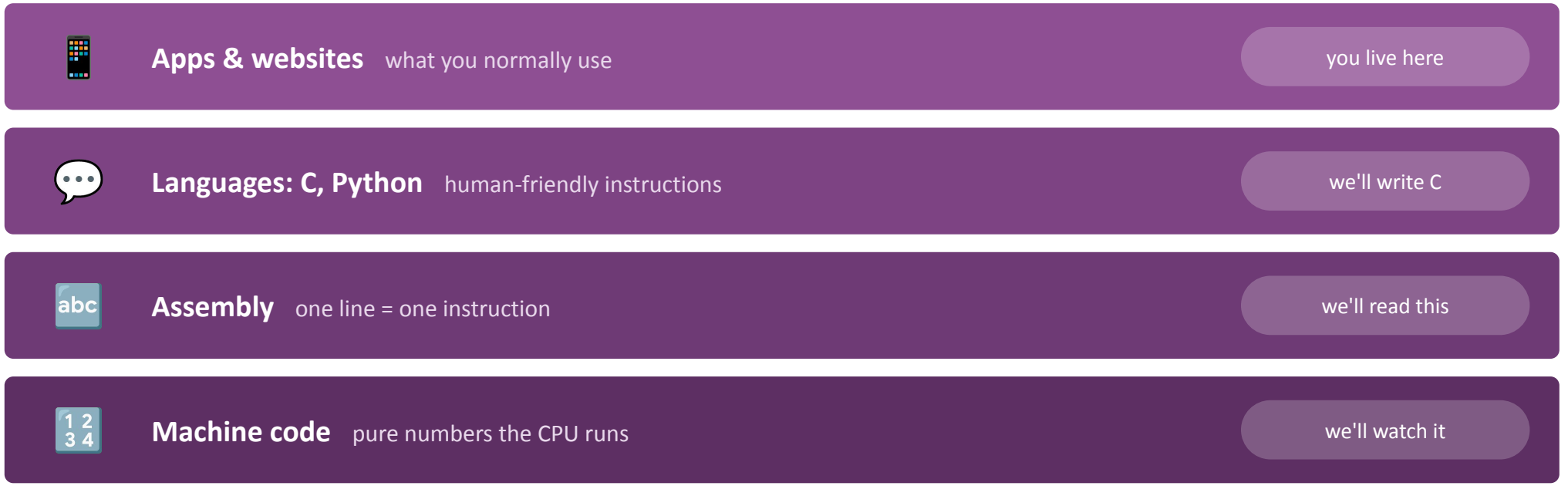

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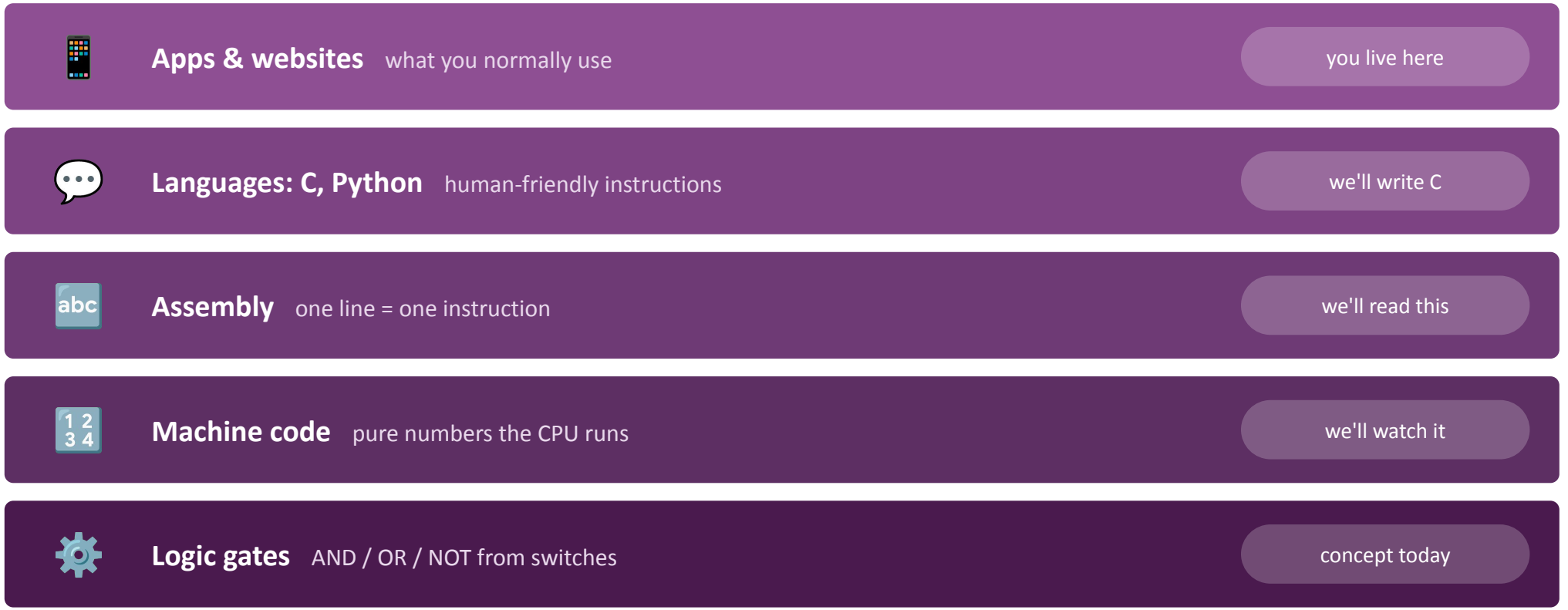


Six rungs from electricity to apps

📱	**Apps & websites** what you normally use	you live here
💬	**Languages: C, Python** human-friendly instructions	we'll write C
🔤	**Assembly** one line = one instruction	we'll read this
🔢	**Machine code** pure numbers the CPU runs	we'll watch it
⚙️	**Logic gates** AND / OR / NOT from switches	concept today
⚡	**Transistors & electricity** physical on/off switches	the bottom rung

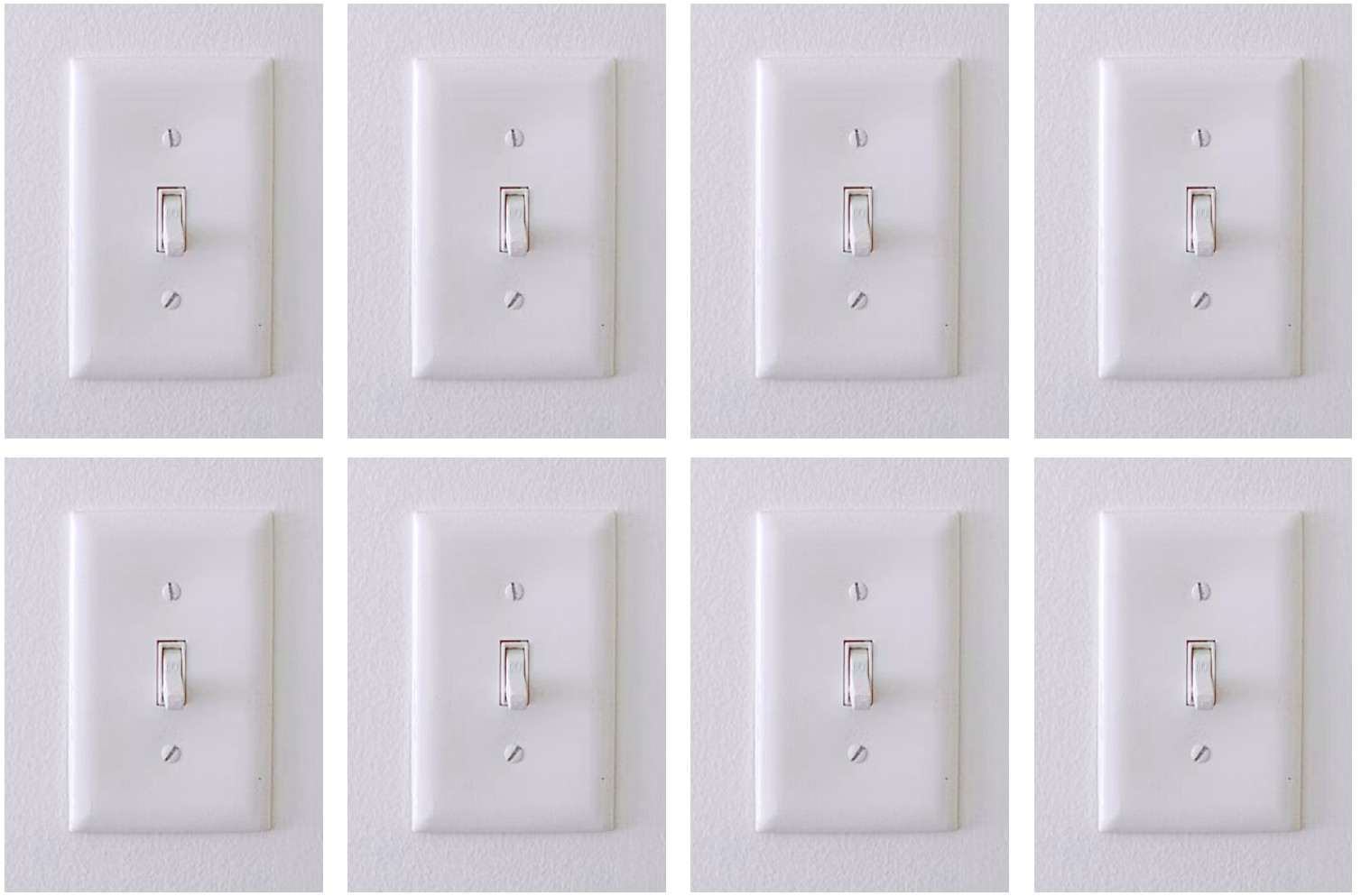
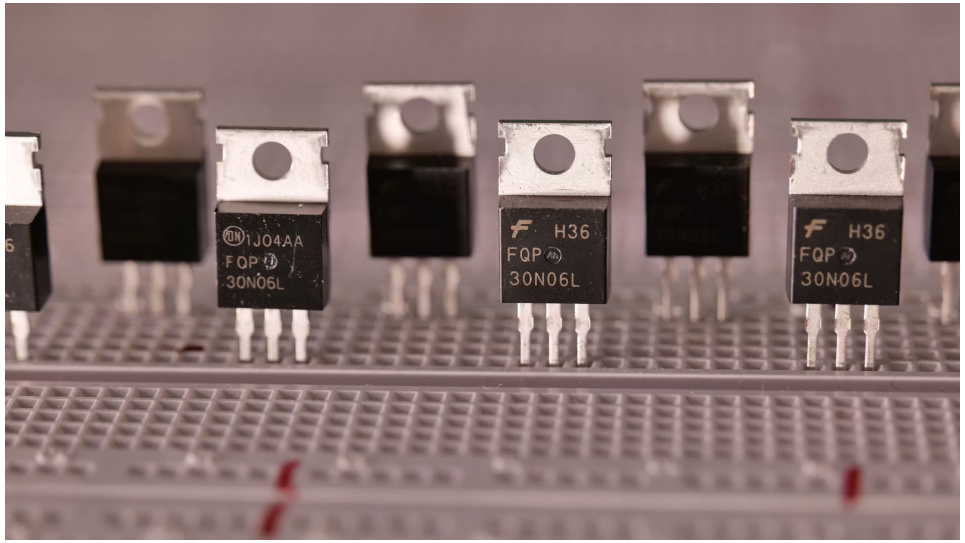
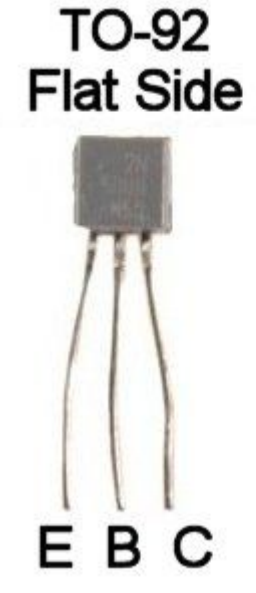


Image source, Unsplash.com





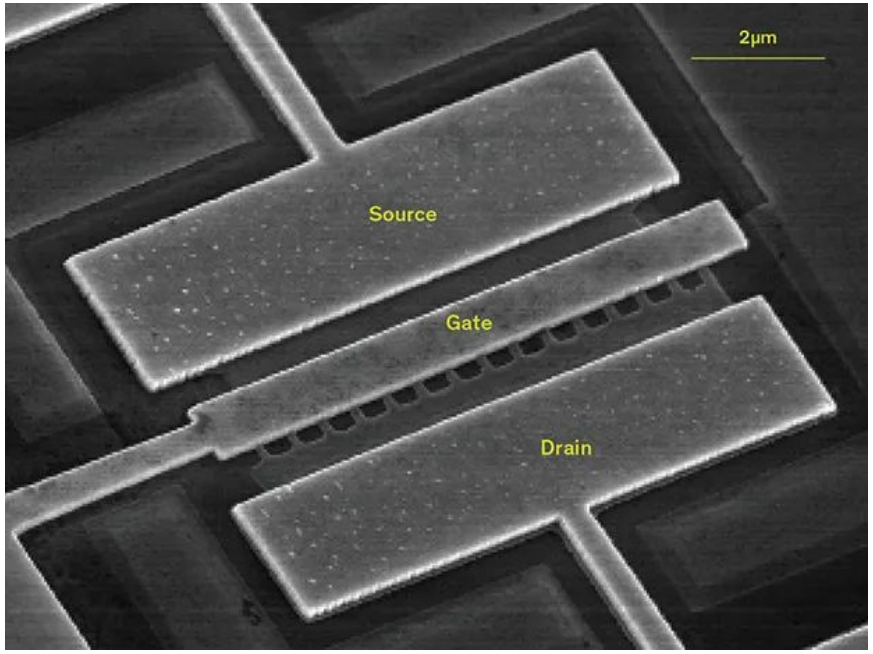


Image sources: IEEE Spectrum, Unsplash.com, EDN News

PART 3 · INSIDE THE BOX

The CPU: fast, and gloriously stupid

A processor does one boring thing billions of times a second. Let's see exactly what.

3

The fetch–execute cycle

1

Fetch

Grab the next instruction. The program counter (pc) remembers which one.

2

Decode

Work out what it means: add? load? jump?

3

Execute

Do it — usually move or add one number.

4

Repeat

Move pc on. Start again. Forever.

That's the whole job. Billions of times a second, it just fetches, decodes, executes, repeats. No cleverness — only speed.

Two places the CPU keeps numbers

Registers: the desk

About 32 tiny slots *inside* the CPU. Lightning fast, almost no room; only what you're working on right now.

```
pc  — which instruction is next
a0  — a number being worked on
t0  — scratch / temporary
sp  — top of the stack
```

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Memory: the filing cabinet

Millions of numbered slots (addresses). Bigger, a little slower. The CPU carries numbers between cabinet and desk all day.

```
.text   — the instructions
.data   — set-up values
stack   — working notes
0x10000 — the 64×64 screen
```

PART 3 · LIVE TOUR

Meet the RV32 Workbench

C code

Write here in a friendly language

Assembly

The CPU's real instructions

Registers

The desk — pc, a0, t0 …

Console

What the program prints

Memory

The filing cabinet

64×64 screen

Pixels drawn from memory

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1 Make it add 2 + 2 (10 mins)

Goal: *watch one instruction run and change a number in front of you.*

1. In the C code box, make the program add 2 + 2 and print it.
2. Look at the Assembly panel: find a line with add or addi.
3. Set the speed to 8/second (or press Step).
4. Watch pc tick forward each step.
5. Keep going until 4 appears in a register, then the Console.

```
int main() {
  print_int(2 + 2);
  return 0;
}
```

✏ **Predict first:** How many Step presses before the 4 appears? Write your guess, then count.

🏋 **Stretch:** Change 2 + 2 to 7 * 6. Set the ISA to RV32IM. Can you spot a mul instruction appear?

2 Muck around with the machine by hand (10m)

Goal: *prove registers and memory are just numbers you can change.*

1. Reset, then Step a few times so registers hold values.
2. Click a register value (try t0) and type 99.
3. Do the same to a slot in the Memory .data area.
4. Press Run and watch your change ripple through.

```
// nothing to type —
// just click values
// and edit them!

t0  =  99
```

🏋️ **Stretch:** Add a line in the Assembly panel, e.g. addi a0, a0, 10. Reset, run — did the answer shift by 10?

3 Watch C become machine code (10m)

Goal: *see one friendly line explode into many tiny CPU steps.*

1. Enter the loop that adds up 1 to 10.
2. On paper, work out 1+2+…+10 first.
3. Run it: did you get 55?
4. Count the Assembly lines: 3 lines of C became many.
5. Slow to 8/sec: watch one register count 1,2,3…

```
int main() {
  int total = 0;
  for (int i=1;i<=10; i++)
    total += i;
  print_int(total);
}
```

✏ **Predict first:** What is 1+2+3+…+10 ? Work it out before you run.

🏋 **Stretch:** Change i <= 10 to i <= 100. New answer? Then add only the even numbers (start at 2, i = i + 2).

4 Paint pixels straight into memory (10m)

Goal: *draw on a real screen by writing numbers — the payoff.*

1. The 64×64 screen lives in memory from 0x10000.

This is some real computers work. We access devices by a number called its memory address (fancy name: memory-mapped I/O)

2. Each pixel is one byte (0–255) = its colour.
3. Fill the screen red with the loop opposite.
4. Try 0x1C green, 0x03 blue, 0xFF white, 0x00 black.

5. Make bands: screen[i] = (i/8)%2 ? 0xE0 : 0x1C;

```
char *fb  = 0x10000;
for (int i=0;i<64*64; i++)
   fb[i] = 0xE0; // red
while(1){}
```

✏ **Predict first:** You never call “draw”. Why does writing numbers change the screen?

🏋 **Stretch:** Use each pixel's position (row = i/64, col = i%64) to invent a gradient or checkerboard. Show your neighbour.

PART 8 · ZOOM OUT

What you just did

You now know

A computer only stores 0s and 1s. Numbers, text and images are all just patterns of them.

You realised

A CPU is a fast, simple machine: fetch, decode, execute. Moving numbers between registers and memory.

You saw that

Code is translated into tiny instructions, and you can read and even write them (but we rarely do this — its too cumbersome!).

In two hours you went from a single switch to running real code and drew a picture by hand-placing numbers in memory. That's the actual machine, with the magic removed.

A computer is not magic.

It's a very fast machine following very simple rules on 0s and 1s. Now you've seen the rules.

Questions?

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